

BASE FITNESS



Long endurance climbs, like those at Maple Canyon, UT, require a high level of Base Fitness. Steve Bechtel on *Toxic Turkey*, 5.13c.

 Janelle Anderson

CHAPTER

5

"At the start of any training regime it's important to get into shape by doing low-intensity, high-volume workouts." — Ben Moon¹³

INTRODUCTION

In nearly every sport, athletes begin their training by establishing a training foundation, which gradually prepares the body for the work to follow in the coming training phases. This base-building phase provides a transition between the active rest period and more intense training activities by gradually exposing the body to a moderate version of the training stress to come. In climbing, this is generally accomplished by climbing at low intensity and long duration, and is often described as "climbing mileage," that is, racking up thousands of vertical feet of climbing. With the critical role that technical skill plays in climbing, these sessions have the added importance of providing an opportunity to learn and refine technique. For grade-obsessed sport climbers, these training sessions also provide an excuse to broaden one's horizons on invigorating, lengthier, moderate climbs.

In route climbing, the difference between success and failure often hinges on the climber's ability to prevent or accommodate mounting fatigue in the muscles of the forearms. To understand how to best train the body to improve these abilities, it is helpful to understand muscle function, metabolic energy pathways, muscle fatigue, and how muscles adapt to training. Therefore, the next several sections will focus on these general physiology topics. If you're not interested in these, skip to the "Aerobic Restoration and Capillarity Training" section on p96.

MUSCLE FUNCTION

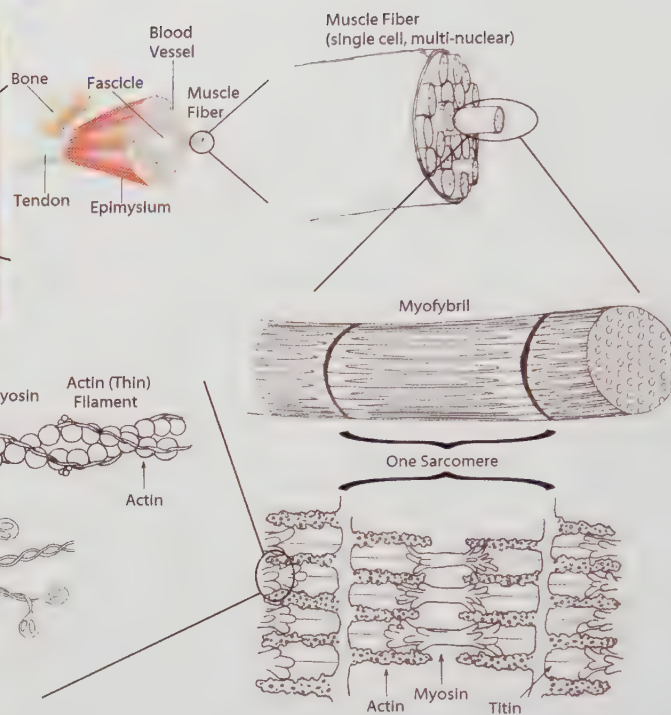
Besides laying a foundation for a training cycle, mileage training establishes the athlete's level of *base-fitness* — the capacity to fuel muscle contractions for sustained activity. Climbing movement requires muscle contractions, and these require energy. In climbing, even *not moving* requires energy (to hang from small handholds). Climbers rely on three metabolic pathways to provide energy for muscular contractions by converting food energy available in the blood (or stored in fat or muscle) into a useable form; these three pathways are discussed starting on p88 in the "Muscle Cell Metabolic Energy Pathways" section. Because of the geometric constraints of the forearms, and the high relative demands placed on them, climbing performance is often limited by the capacity of these metabolic pathways in the forearms themselves. The limitations of these pathways are manifested as a forearm "pump."

Muscles create joint movement, so they are fundamental to sports performance. Skeletal muscles undergo three types of contractions:

- *Concentric* (muscle shortens during contraction)
- *Eccentric* (muscle lengthens during contraction)
- *Isometric* (muscle length remains fixed during contraction)

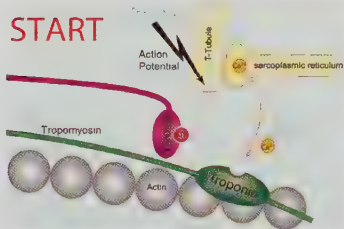
Isometric contractions are used by the forearm muscles to grasp handholds, and are thus critical in climbing. Isometric grip strength and endurance is often a limiting factor for many climbers and on many climbs.

Skeletal muscles are composed of *blood vessels* and *muscle fibers*, and are attached by tendons to the bones (see Figure on the next page). A muscle fiber is actually a long, tubular cell composed of numerous *myofibrils*, which are the smallest individual contractile units in the muscle. Myofibrils are segmented into



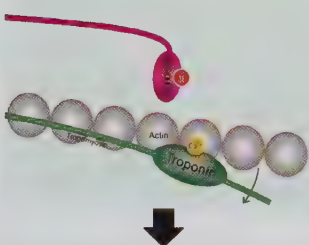
Human skeletal muscle (adapted from "Skeletal Muscle" © 2005 under a Creative Commons Attribution-ShareAlike license: creativecommons.org/licenses/by-sa/3.0/ photo: Brendan Nicholson

START

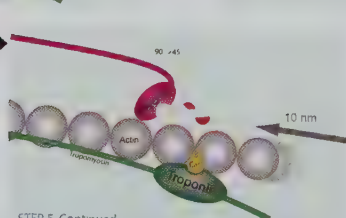
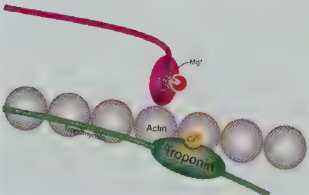


If Action Potential is Received

STEP 4

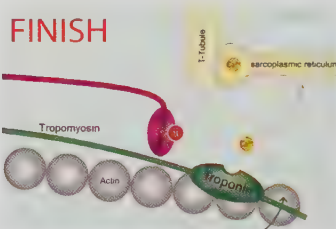


STEP 5



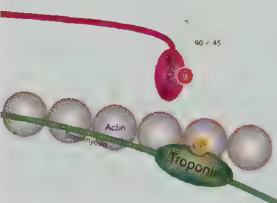
STEP 5, Continued

FINISH

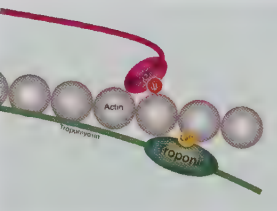


If Ca^{2+} and ATP are Depleted

STEP 7



STEP 6



Muscle Function Glossary:

- ADP - adenosine diphosphate
- AP - action potential
- ATP - adenosine triphosphate
- Ca^{2+} - calcium ion(s)
- H^+ - hydrogen ion(s), or "proton"
- Mg^{2+} - magnesium ion(s)
- MSS - maximum steady state
- PCr - creatine phosphatase
- P_i - inorganic phosphate
- ROS - reactive oxygen species
- SR - sarcoplasmic reticulum

Illustration of muscle contraction (adapted from "Molecular Mechanism of Muscle Function" © 2007 by Hank van Helvete, used under a Creative Commons Attribution-ShareAlike license: creativecommons.org/licenses/by-sa/3.0/

Muscle Contractions

The process of initiating and sustaining muscle contractions is complex, with many moving parts. However, this complexity provides numerous controls to delay fatigue and prevent permanent muscle damage. Muscles contract in several steps (refer to the Figure on the previous page):

1. A contraction command originates in the brain, generating an electrical signal called an *action potential* (AP) that is propagated to the muscle by a *motor neuron*.
2. The motor neuron releases a molecule, *acetylcholine*, which travels across the *neuromuscular junction* to the muscle cell, thus transmitting the AP to the muscle.
3. The AP propagates through the muscle cell along *transverse tubules* (T-tubules) to the *sarcoplasmic reticulum* (SR), which contains stored *calcium ions* (Ca^{2+}). The AP stimulates release of Ca^{2+} ions from the SR, ions which then diffuse into the *cytoplasm* (the fluid within the muscle cells) and surround the *myofibrils* (made up of actin and myosin).
4. The Ca^{2+} ions control the myofibrils and stimulate contractions through a protein called *tropomyosin* that resides on the actin thin filaments. Normally, the tropomyosin prevents myosin from binding to the actin, but when Ca^{2+} ions are released from the SR, they bind to *troponin C* receptors on the tropomyosin, which moves it out of the way.
5. The myosin heads (which are preloaded with *adenosine triphosphate* [ATP] — the primary energy source for muscle movement) are now able to attach to the recently uncovered binding sites on the actin thin filaments. This process requires *ATP hydrolysis* (ATPase), in which the ATP releases energy while splitting into *adenosine diphosphate* (ADP), *inorganic phosphate* (P_i), and *hydrogen ions* (H^+), all of which are important elements of fatigue. When the P_i and ADP are released, the myosin head rotates, pulling on and moving the actin approximately 10 nanometers.
6. ATP available in the cytoplasm then binds to the myosin head, replacing the ADP and releasing the myosin head from the actin binding site.
7. The iterative process of *cross-bridging cycling* is thus initiated (steps 5 and 6). As long as Ca^{2+} ions and ATP are present in the cytoplasm, the myosin heads will repeatedly attach, pull, detach, and re-attach as they inch along the actin fiber like a ratchet, shortening the sarcomere.
8. Throughout this process, Ca^{2+} ions are actively pumped back into the SR (a process which also requires ATP). The SR will continue to release Ca^{2+} ions as long as an AP is present, and the contraction will persist. When the AP is turned off, the Ca^{2+} pumps will clear the cytoplasm of Ca^{2+} ions, and the tropomyosin will move back into the blocking position, preventing further cross-bridge cycling.

sarcomeres, forming a long chain from one end of the muscle to the other. The sarcomeres are the contracting elements of the cell, shortening, lengthening, or resisting motion, as directed by the nervous system through *motor neurons*. Sarcomeres contain bundles of long, thin proteins called *actin* (thin filament) and *myosin* (thick filament) that interact in a complex electrochemical process to create contractions.

This electrochemical process begins in the central nervous system and acts on the musculoskeletal system, but is supported by many other systems that provide numerous chemical ingredients. The process is complex, involving many components in several steps. This complexity lends robustness, with numerous controls to delay fatigue and prevent permanent

muscle damage. A muscle contraction is initiated by an *action potential* (AP) from the nervous system that is transmitted to the motor neurons, which activate the myofibrils and sarcomeres. This initiates the process of *cross-bridge cycling* in which *adenosine triphosphate* (ATP) provides energy for myosin thick filaments to attach to actin thin filaments and pull them along, 10 nanometers at a time. In the case of a concentric contraction, this ratcheting process shortens the sarcomere, and thus the entire myofibril chain; this then shortens the muscle. As long as the nervous system requests it, and the necessary metabolic components (such as ATP) are available, the contraction will continue. More details are provided in the above sidebar.

MUSCLE CELL METABOLIC ENERGY PATHWAYS

Large quantities of ATP are needed to power sustained muscle activity, and the human body has robust and redundant processes to provide it. ATP is produced from biological fuels, including sugar (carbohydrates), amino acids (protein), and fatty acids (fat). Sugar, in the form of blood glucose, is by far the preferred fuel. Digested carbohydrates enter the bloodstream as glucose, an energy-dense sugar, which is either used immediately, or converted to *glycogen* for storage in the muscle cells through the *glycogenesis* process.

GLYCOLYTIC PATHWAY

Glucose and glycogen are eventually converted to ATP and *pyruvate* molecules through the complex process of glycolysis (the heart of the glycolytic energy pathway). Glycolysis is a ten-step process with several variations, but generally, it is an anaerobic process (not requiring oxygen) that quickly produces a modest amount of ATP: two ATP molecules per molecule of glucose. A consequence of glycolysis is that, in addition to ATP, it produces a number of by-products that contribute to muscle fatigue, including *hydrogen ions* (H^+) and the molecule *nicotinamide adenine dinucleotide* (NADH). However, when possible, these by-products are consumed by *aerobic respiration* to produce even more ATP while simultaneously recycling these waste products. This glycolytic energy pathway is a key contributor to performance on power-endurance climbs (those with sustained sections of very difficult moves). **Chapter Eight: Power-Endurance** will introduce training methods to improve glycolytic metabolism.

AEROBIC RESPIRATION PATHWAY

Aerobic respiration occurs inside the *mitochondria* (an energy-supplying organelle found in many cells), and is the general process of oxidizing pyruvate in the “Krebs” cycle. Aerobic respiration picks up where glycolysis leaves off, by using the products of glycolysis as its primary fuel (in addition to oxygen). Aerobic respiration is much more productive than glycolysis, producing roughly 28 ATP molecules per molecule of glucose, above and beyond the two generated during glycolysis. More importantly, the ATP is produced largely by metabolizing the waste products NADH, H^+ , and inorganic phosphate (P_i) through a process called *oxidative phosphorylation*. During moderate-intensity, steady-state exercise, aerobic respiration completely consumes these otherwise harmful *metabolites* (harmful by-products of cellular energy me-

tabolism), allowing exercise to continue as long as glucose is available. The *Maximum Steady State* (MSS – described in detail in “Capillarization and the Maximum Steady State”) is the highest level of intensity at which a muscle can continue to function aerobically, and thus, indefinitely. Climbing *below* the MSS typically does not cause a significant forearm pump and can be sustained for many minutes. Climbing continuously *above* the MSS will result in a debilitating pump within a few minutes. This process makes continuous endurance feats (like running a marathon) possible. The aerobic energy pathway is used continuously, in all aspects of climbing, to provide ATP and recycle waste, but can be especially critical to success during onsights, very long, sustained redpoints, and long multipitch climbs.

PHOSPHAGEN PATHWAY

The *phosphagen* system is the third and final method for producing ATP for muscle activity. It delivers ATP almost immediately, and provides the supply needed for the first few seconds of a muscle contraction. ATP is produced in a reaction called *creatine kinase*, in which readily available creatine phosphate (PCr), ADP, and H^+ react in one step to produce ATP and creatine. The stores of PCr are quickly depleted, however, and ATP production must be replaced by aerobic respiration or glycolysis. (Creatine supplementation is used by some athletes because it boosts phosphagenic ATP production, thus improving high-intensity, short-term [<10 sec] performance, but it has no known enhancing effect on longer-term efforts.) By enabling maximum force production, the phosphagen pathway is critical to bouldering, and to performing bouldery moves on routes.

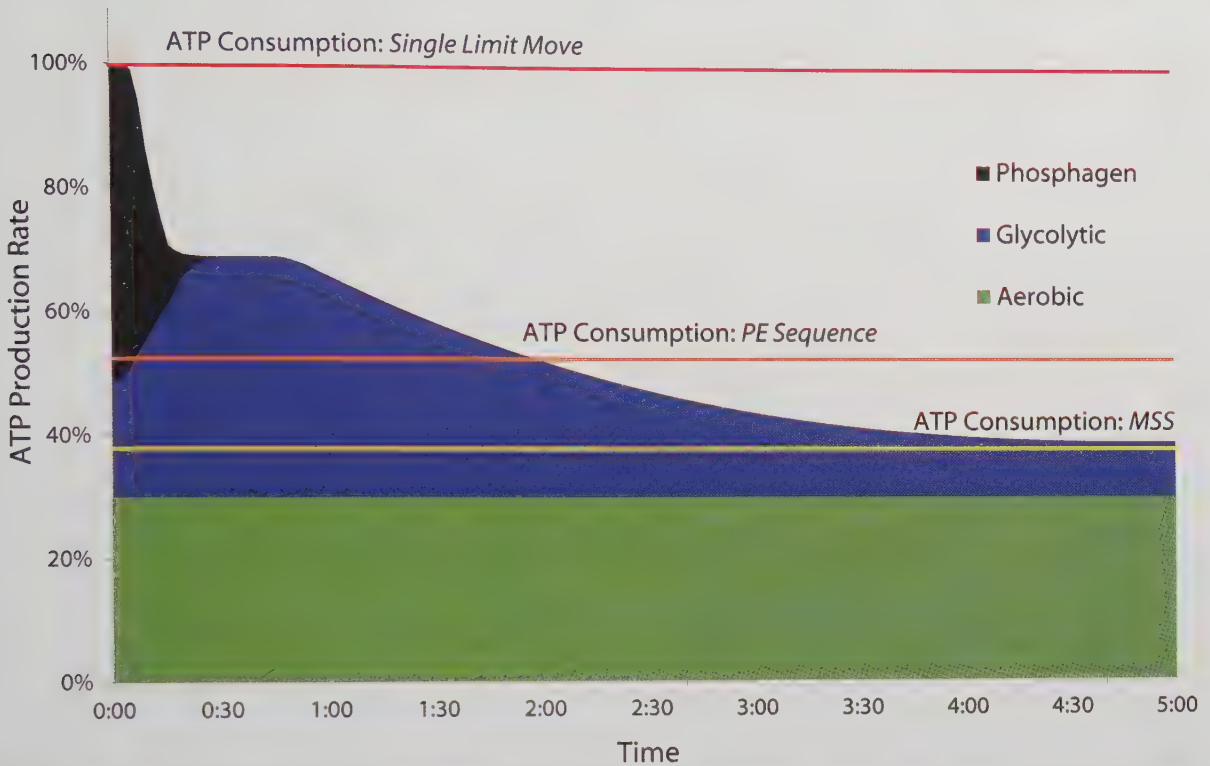
ATP: ENERGY FOR MUSCLE CONTRACTIONS

Before discussing the pros and cons of each metabolic energy pathway, it is necessary to briefly mention how ATP is used. Once ATP is produced in the muscle cells by one of the three metabolic systems, it is available for use by the myosin heads to power muscle contrac-

Metabolic Pathway	ATP Production Rate (mmol ATP/s)	
	Anaerobic	
Phosphagen		2.4
Glycolytic		1.3
Aerobic		
Aerobic Respiration (carbohydrate)		0.7
Aerobic Respiration (fat)		0.3

ATP Production via the various Metabolic Energy Pathways.⁸

Relative Contributions of the Metabolic Energy Pathways to ATP Production



Relative Contributions of the Metabolic Energy Pathways to ATP Production. All three systems function simultaneously, at first. The Phosphagen Pathway is necessary for executing the most difficult moves, but it can only be sustained for a dozen or so seconds. The Glycolytic Pathway produces ATP rapidly, but can only sustain that production for one to three minutes. Aerobic respiration can produce ATP indefinitely, but at a slower rate. Various climbing activities consume ATP at different rates (shown notionally). Limit moves require tremendous amounts of ATP, so can only be performed until the Phosphagen Pathway is exhausted, which takes only a few seconds. Power-endurance sequences can be performed until the Glycolytic Pathway is exhausted, which takes another two minutes or so. Climbing at or below the Maximum Steady State can continue for dozens of minutes.

tions. (ATP has other uses as well, including transporting molecules across cell membranes and powering certain steps of glycolysis and aerobic respiration.) The mechanical work performed by the myosin head as it tugs on the actin filament is powered by the release of energy that accompanies *ATP hydrolysis* (aka ATPase). ATP hydrolysis is a reaction between ATP and water that yields the by-products P_i , H^+ , and *adenosine diphosphate* (ADP), along with the release of energy. Muscles cannot function without ATP hydrolysis, yet the three by-products are largely responsible for muscle fatigue. Therefore, the rate at which ATP is consumed (and these waste products are produced), along with the ability of the muscle cells to deal with these waste products, largely determines how long muscle activity can continue.

This is why the metabolic energy pathways are critical: because each produces ATP and handles the associated waste products differently. However, keep in mind that all three pathways function simultaneous-

ly, but to varying degrees, depending on the climbing intensity. The relative contribution of each pathway to the total supply of ATP determines the sustainability of the climbing activity. The figure above illustrates how each pathway contributes to ATP production.

Mitochondrial aerobic respiration is the body's preferred choice for producing ATP during exercise. It consumes all of the by-products of ATP hydrolysis at the same rate as they are produced; therefore, these detrimental metabolites do not accumulate in the blood and muscle cells, they do not interfere with muscle function, and thus fatigue does not ensue. Unfortunately, aerobic respiration is also the *slowest* form of ATP production (see Table on the previous page). Therefore, as climbing intensity increases, the muscles consume ATP faster than it is produced *aerobically*. In response, the cells produce ATP faster through increased use of the unsustainable *anaerobic* glycolytic energy pathway (initiating the countdown to fatigue).

Lactic Acid...Not Guilty!

For nearly 80 years, the mainstream explanation for muscle fatigue was the familiar “burn” of lactic acid accumulation (or lactic acidosis) in the muscle cells. While exercise is closely correlated with a decrease in blood pH (making the blood more acidic), no evidence for a cause-and-effect relationship between lactic acidosis and muscle fatigue has ever been discovered.⁸ The lactic acid *theory* of fatigue has since been roundly rejected by medical science, and technically, “lactic acid” is not produced by the human body — rather, sodium lactate (or just lactate) is. Lactate is a natural by-product of glycolysis, and it does not cause acidosis, or muscle fatigue.¹ Acidosis is caused by the excess hydrogen ions (H^+) released during ATP hydrolysis (the breakdown of ATP into free energy, ADP, P_i , and H^+). While lactate production does correlate with increased acidity in the blood, the lactate is a *result* of the acidity, not a cause of it, and the lactate actually helps to buffer (remove) excess H^+ , reducing acidity and permitting exercise to continue longer than it would otherwise.

Despite the fact that acidosis has been closely correlated with muscle fatigue, it’s currently not clear how or if acidosis causes fatigue.¹⁰ One hypothesis supported by limited experimental evidence is that lowered pH interferes with the binding of Ca^{2+} to troponin C on the actin thin filament, thus reducing force production (but only slightly).¹⁰ Acidity has also been implicated in the slowing of relaxation that occurs with muscle fatigue.¹

Inorganic phosphate (P_i), however, is also a by-product of ATP hydrolysis, and it is well-known to cause muscle fatigue and reduce force production. Increased concentrations of P_i encourage the muscle cells to use the weaker of two possible actin-myosin cross-bridge states, thus reducing the force output of the muscle.¹⁰ Further, increased P_i levels inhibit the release of Ca^{2+} from the sarcoplasmic reticulum (needed for cross-bridging between the actin and myosin filaments). Therefore, the presence of excess P_i inhibits force production in two ways: 1) It reduces the number of actin-myosin cross-bridges, and 2) It reduces the strength of those bridges.



“Excess Inorganic Phosphate Bath” just doesn’t have the same ring to it. Rachel Avallone pumping out on *Lactic Acid Bath*, 5.12d, at the New River Gorge, WV.  Dan Brayack

While glycolysis is less *efficient* at producing ATP (only two ATP per molecule of glucose) than aerobic respiration, it is much *faster*. Glycolysis produces ATP at nearly twice the rate of aerobic respiration, doubling the amount of ATP available but also doubling the production of fatigue-causing waste products. Further, glycolysis itself produces excess protons (H^+) in addition to those produced by ATP hydrolysis. The mitochondria still recycle as much of these harmful by-products as they can through aerobic respiration, but these efforts will eventually be outpaced by production. If physical activity continues, these waste products will accumulate and cause muscle fatigue.

MUSCLE FATIGUE

Muscle fatigue can result in reduced force production, reduced contraction speed, reduced relaxation speed, or a combination of all three. All of these manifestations contribute to an overall decline in power: the muscle's ability to produce a large force quickly. In

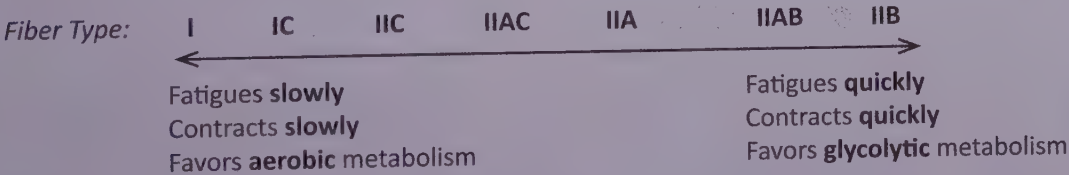
rock climbing, muscular power is fundamental to performing dynamic movements, and thus to negotiating difficult cruxes. When muscles perform continuous maximum contractions (such as gripping a tenuous handhold), force production declines rapidly (within five to ten seconds), but recovers rapidly as well (one to two seconds). Of course, this exertion/recovery cycle cannot proceed indefinitely, and fatigue will build within a few repetitions.

Sustained rock climbs demand that climbing muscles (in the forearms especially) contract and relax repeatedly, but often at considerably less than maximum force. Running and cycling tax muscles similarly. The force required and the *duty cycle* of these contractions are critical factors in determining how long the effort can be sustained. The duty cycle is the ratio of contraction time to rest time (a two-second contraction with a one-second rest is a duty cycle of 2:1). Running and cycling have low force requirements and low duty cycles (less than one), so these activities

Muscle Fiber Types

Each muscle is composed of many *motor units*, each consisting of many individual muscle fibers. Currently, there are seven known muscle-fiber types in humans, and these form a spectrum ranging from high-fatigue-resistance fibers that contract slowly, to low-fatigue-resistance fibers that contract quickly. Classic models only recognized three fiber types, and for pragmatic purposes, the four newcomers are still grouped with the original three categories most of the time:

- *Type I* fibers are *slow twitch* (meaning they contract slowly) but they are well-equipped for aerobic respiration, allowing them to sustain longer-duration activities. These fibers take two to three times as long to contract as the so-called *fast-twitch* fibers.
- *Type IIa* are one of two primary types of *fast-twitch* fibers (meaning they contract quickly). Type IIa fibers contract relatively quickly, but are also somewhat fatigue-resistant.
- *Type IIb* (aka *Type IIx/d*) are the fastest contracting fibers, but they fatigue quickly. De-training can cause the conversion of Type IIa fibers to Type IIb.



The relative distribution of these muscle-fiber types varies by individual and muscle group (and this distribution *can* vary between muscle group), so to some extent an athlete's propensity toward endurance or power-oriented activities is genetically predetermined. Fortunately for athletes, it is possible to train intermediate-twitch Type IIa fibers to behave more like faster-twitch Type IIb fibers and vice versa (it also may be possible to convert Type I to Type II, but clinical evidence is lacking). Furthermore, it is possible to train Type I fibers to contract more quickly while retaining other "slow-twitch" characteristics (such as fatigue resistance). Therefore, while you may be predisposed toward a particular type of climbing (i.e., power moves versus endurance routes), there is significant potential to train your muscles to behave in a way that suits your individual goals, regardless of genetics.⁹

Causes of Muscle Fatigue

Given the complex system of systems involved in muscle contractions, there are numerous failure points that can cause fatigue. Several fatigue modes have been discovered, but it is difficult to implicate a specific mode (or modes) of fatigue for a given activity.¹ Exercise variables such as load, duty cycle, and duration factor into how fatigue develops. In cyclic activities (such as climbing), fatigue develops in three phases as shown in the Figure below. In phase one there is a rapid decline in force production of 10 to 20 percent, caused by reduced effectiveness in cross-bridge cycling as PCr is depleted. Phase two incurs a more gradual decline in force, and its length depends on the effectiveness of aerobic respiration. Finally, in phase three, muscle performance declines rapidly as Ca^{2+} function is impaired by accumulating metabolites. The most relevant fatigue modes for climbing are:

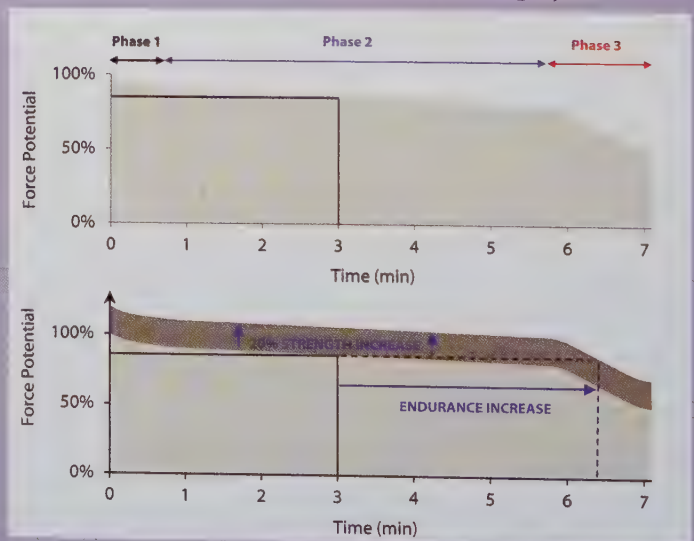
ATP Depletion: During cruxy climbing, fast-twitch fibers consume ATP rapidly. The phosphagen system supplies this ATP for the first few seconds, but as PCr is depleted, the cells must turn to slower glycolysis, which is soon outpaced by demand, and ATP levels decline in the cytoplasm. When the SR senses this shortage, Ca^{2+} ion release is intentionally reduced to prevent cell damage, and the muscle can't generate as much force (dietary creatine delays this fatigue by one to two seconds, but the associated weight gain may be detrimental for climbers).

Metabolite Accumulation: During power-endurance climbing efforts, fast-twitch fibers are recruited and loaded cyclically, and the metabolites P_i , H^+ , ADP, and Mg^{2+} (magnesium ions) accumulate quickly. Other metabolites called *reactive oxygen species* (ROS aka *free radicals*) arise very gradually from mitochondrial respiration. Generally, all of these metabolites prevent Ca^{2+} ions from doing their job. Therefore, a climber's ability to deal with accumulating metabolites is critical to performance. Mitochondria perform this cleanup function in the cells, so any training that promotes mitochondrial performance is very beneficial.

Glycogen Depletion: It's possible to cause fatigue by depleting the body's glycogen reserves, but it takes a long time, so the exercise would have to be relatively easy. Therefore, it is difficult to *climb* into "glycogen debt" (although the Nose-in-a-Day or 24 Hours of Horseshoe Hell would be potential cases). Limit climbs will incur faster-developing fatigue modes (such as metabolite accumulation) that cause failure well before glycogen runs out. However, glycogen depletion *does* contribute to fatigue in higher-intensity activity such as cruxy climbing, by reducing force output over time. Reduced Ca^{2+} ion output by the SR is the likely culprit.

Neuromuscular Action Potential Disruption: The electrochemical channels that are used to stimulate muscle contractions can de-polarize with use (like a worn-out magnet). This fatigue mode mostly affects continuous maximum contractions. Muscles can overcome this, but only if thoroughly stimulated (through use). Therefore, a warm-up that includes significant, near-maximal exercises is required to get the muscles "firing on all cylinders."

Development of muscle fatigue showing that as the muscle is fatigued by cyclic sub-maximal loading, its potential to produce a single maximum-force contraction declines.¹ For example, if the fatiguing activity were a pumpy climb, the top figure indicates that three minutes into the climb, the climber would only be able to perform moves requiring about 80 percent or less of her maximum strength. The bottom figure shows the effect of increasing the muscle's maximum strength by 20 percent. The same hold that required 80 percent of maximum strength can now be held with only 65 percent of maximum strength, so it can still be used six or more minutes into the climb. The apparent effect is an increase in endurance.



can be sustained for long periods. Climbing, on the other hand, will always have a duty cycle of greater than one (assuming that one hand contacts the rock at all times). Sub-maximal, repeated contractions can be sustained much longer than continuous maximal contractions, but once fatigued by this cyclic loading, recovery takes longer: five to ten minutes for initial recovery, and as much as 30 minutes to two hours to regain maximum strength. In general, the lower the load, the longer the effort can be sustained, but also the longer recovery will take. This is known as “low-frequency fatigue” and can linger for days. Therefore, long bouts of fatiguing, low-intensity climbing can dig a deep hole that requires days to “climb” out of.

Fatigue at the muscle cellular level arises through the accumulation of by-products from ATP metabolism. These metabolites interfere with numerous steps in the muscle contraction process, as described in the adjacent sidebar.

The three processes for producing ATP are not mutually exclusive — their relative contribution to a given muscle contraction or exercise effort varies as contractions progress and metabolic conditions change within the cell. The muscle fiber type also affects the relative reliance on any one system. So-called *fast-twitch* fibers use ATP more quickly than *slow-twitch* fibers; therefore, they lean heavily on glycolysis to produce ATP very quickly. Slow-twitch fibers have higher concentrations of mitochondria, and are better at utilizing aerobic respiration; therefore, slow-twitch fibers are *fatigue-resistant*. As a consequence, when a climber attempts difficult moves requiring high muscular force and/or power output, the more powerful fast-twitch muscle fibers will be recruited for the task. However, these fibers lack sufficient mitochondrial aerobic respiration, so fatigue will develop quickly. These muscle characteristics have profound implications on how climbers should best train for their sport, and they impact all physiological aspects of climbing performance: strength, power, and endurance.

MUSCLE CHARACTERISTICS AND THEIR IMPLICATIONS FOR CLIMBING TRAINING

Muscle cells have several fatigue mechanisms that are triggered by different conditions within the cells. Therefore, the fatigue mechanism is specific to the activity being performed.³ So when training to increase fatigue *resistance* in climbing, it is critical to create the right type of fatigue in order to yield the desired adaptations. That is, the training must be *climbing specific* in order to develop muscular endurance that is rele-

vant to climbing. For best results, the training should be tailored to mimic specific aspects of the *goal route*, such as the muscle loads (hold type), duty cycle, and total-effort duration. Specific training will achieve the most-applicable improvements to muscle performance, best preparing the climber for his goal route. Another effect of the muscles’ versatility is that climbers can improve their endurance by training for strength. Relatively high-force contractions recruit more fast-twitch fibers, which produce more force but fatigue more quickly. By training to increase maximum strength, a given climbing move will require a relatively lower percentage of that maximum strength, and thus relatively fewer fast-twitch fibers.¹ Instead, the more fatigue-resistant slow-twitch fibers can support more of the workload, resulting in less fatigue. Over the course of an entire climb, the increased use of slow-twitch fibers delays the accumulation of fatigue significantly. In short, strength training improves muscular endurance, which is one reason it should be central to any climbing training program.

Another important ramification for training is the ability of all fibers (including fast-twitch) to respond to endurance training by adding mitochondria. This adaptation does not impair the fibers’ fast-twitch performance, but does allow them to perform longer before fatiguing.⁹ Therefore, climbers should train to increase the strength and quantity of fast-twitch fibers, while also improving their fatigue resistance. Thus, high-intensity power training *and* endurance training are both beneficial and necessary for optimizing overall climbing performance.

Finally, it is essential to emphasize the critical importance of mitochondria in the performance of all types of muscle fibers (slow and fast). Regardless of the duration or intensity of the activity (whether bouldering or enduro-climbing), mitochondria are involved in clearing waste and efficiently providing ATP. Their presence and effectiveness will prolong optimal muscular performance during any climbing activity. The quantity and capacity of mitochondria can be increased by moderate-intensity “threshold” training, which will benefit all aspects of climbing performance.⁹

THE FOREARM PUMP

One unfortunate but consequential feature of skeletal muscles is that high-force contractions restrict blood flow to the contracting muscle. Muscles are infused with small capillaries to supply necessary fuel, but as the sarcomeres contract, they also widen (easily ob-



served in a flexing bicep), constricting the surrounding capillaries. This *blood-flow occlusion* has long been suspected by climbers, and recent research has verified it. Studies have shown that circulation in the forearms is severely reduced or eliminated during intense contractions (50 to 70 percent of maximum strength or greater).^{2, 3, 5, 6} To make matters worse, one study has shown that raising the arms above the head during moderate, repetitive isometric hand contractions reduces local blood pressure and hastens fatigue (suggesting climbers should alternate shaking out above and below their heads to promote blood flow in and out of the forearm muscles).¹¹ Blood flow provides the essential fuels of oxygen, glucose, and ATP for aerobic and anaerobic respiration, and it removes the fatigue-inducing metabolic waste products, along with CO_2 .^{1, 3} Muscles fatigue more rapidly when blood flow is occluded, whether the fatiguing exercises are high-intensity maximal contractions (cruxy climbing) or endurance activities (enduro-climbing).

In climbing, blood-flow occlusion is especially problematic. During difficult climbing, blood flow to the forearm muscles is severely reduced. Furthermore, to handle the strenuous moves, less efficient metabolic pathways are employed to rapidly produce ATP. As a result, metabolites from ATP hydrolysis and glycolysis quickly accumulate, while much-needed

High force contractions in the forearm muscles occlude blood flow, eventually resulting in the dreaded forearm pump. Janelle Anderson on *Jesus Wept*, 5.12d, Red River Gorge, KY.  Mike Anderson

blood glucose and oxygen “reinforcements” are prevented from entering the muscle tissue. This problem is compounded by the muscle’s growing inability to *stop* contracting. The myosin heads require ATP to release from the actin binding sites (step six of muscle contraction), so as ATP is depleted, some fibers may be unable to relax, perpetuating blood-flow occlusion. This vicious cycle is well-known to climbers as the dreaded forearm pump, which reduces strength, speed of contraction/release, and coordination, all greatly impairing climbing performance.

Climbing typically loads the forearm muscles cyclically with five-to-ten-second periods of high-intensity, blood-occluding isometric contractions, separated by brief (~ one second) rests with resumed blood flow (shaking out permits a longer rest). Therefore, a long, sustained pitch of hard climbing can be thought of as an ebb and flow between mounting muscle fatigue and recovery. Each contraction draws the climber closer to failure, while each release provides a brief retreat from it. Assuming the climber can do the moves, success on a difficult climb may then depend on his ability to repeatedly, briefly, and consistently recover between handholds.

Muscular Recovery from Fatiguing Exercise

The recovery of muscle performance following fatiguing exercise has interesting characteristics that inform climbing training. When muscles perform short-duration, maximum-force contractions (like repetitive weightlifting), strength recovers very quickly (almost completely in only a couple of seconds). Generally, the longer the length of continuous contraction, the longer recovery takes. Fatigue caused by repetitive, sub-maximal contractions, like those required for route climbing, requires much longer recovery in general, but also depends on the level of effort and duration of activity. Relatively easy moves that can be sustained longer require more recovery time than harder moves that cause fatigue quickly. This is why climbers typically rest longer after pumping off a long route (with many, relatively easier moves) than after failing on a boulder problem (with fewer, more difficult moves).

While a substantial amount of recovery occurs in the minutes and hours immediately following fatiguing climbing, there is a component of fatigue that lingers for many hours or days. This fatigue is not a result of reduced ATP or the accumulation of metabolites (these return to normal levels in 15 to 60 minutes). Instead, it results from a breakdown in the neuromuscular excitation process that stimulates the SR to release Ca^{2+} (needed for cross-bridge cycling). This impairment persists for many hours or days after intense exercise.¹ These characteristics of muscle recovery can be used to plan rest cycles to optimize climbing performance. At the Red River Gorge, it is not unheard of to rest for up to two hours between attempts on long, sustained routes, whereas the roped-bouldering routes of Wyoming's Wild Iris may require only 20 minutes of rest between burns. Furthermore, more rest may be required between climbing days filled with long, pumpy climbing.

RECOVERING FROM A FOREARM PUMP

Recovery from fatigue is severely hindered by occluded blood flow. Regardless of the intensity and duration of the climbing activity, it can be sustained longer and stronger with improved blood flow. Blood flow to the forearms during a climb can be improved through

two primary mechanisms: The first is to reduce the percentage of maximum contraction used to complete the move. This can further be accomplished in two ways: First, by increasing the climber's maximum finger strength ([Chapter Six: Strength](#)) so that, for any given move, a lower percentage of maximum effort is required (See Figure on p92); and second, by improving grip control ([Chapter Three: Skill Development](#)) so that the bare minimum amount of force is applied to each hold.

The second primary mechanism for increasing blood flow to the forearms is to improve the capillary network throughout the forearm muscles, or *capillarization*. Oxygen enters the muscle fibers passively through diffusion. Therefore, blood must be positioned very near the muscle fibers for the best transportation of this critical fuel. Through proper training, it is possible to increase the quantity of capillaries, as well as their size, thus increasing blood flow, oxygen delivery, and waste removal.

CAPILLARIZATION AND THE MAXIMUM STEADY STATE

Recall that the aerobic energy pathway can function indefinitely, as long as glucose is available. The highest level of intensity at which a muscle can continue to function aerobically, and thus, indefinitely, is called the *Maximum Steady State* (MSS; aka aerobic threshold, anaerobic threshold, or lactic threshold). The level of capillarization within a muscle is a key determiner of the MSS. Climbing at an intensity level above the MSS relies sufficiently on anaerobic respiration (glycolytic and/or phosphagen) that such effort becomes unsustainable, ultimately resulting in fatigue and eventual muscle failure. In endurance sports, it is often said that the MSS is the singlemost important determinant of performance, and so the majority of training activities are geared toward improving it.

In climbing, the importance of the MSS varies depending on the goals of the climber, i.e., the characteristics of the routes the climber attempts. The more sustained the climb, the more important MSS will be, but as described previously, *all climbing activities will benefit from improved capacity for aerobic respiration within the muscle fibers*. However, this characteristic must be specific to the muscles used (mostly the forearms) for it to be relevant to climbing. The whole body's systemic capacity for aerobic respiration is

largely irrelevant in rock climbing, while the aerobic capacity of the muscle cells within the forearm is of utmost importance. Therefore, training must be properly tailored to the demands of climbing to improve local capillarity of the forearms in order to raise their MSS. Fortunately, this type of training has long-lasting effects, and investments in improving capillarity will pay dividends for years.

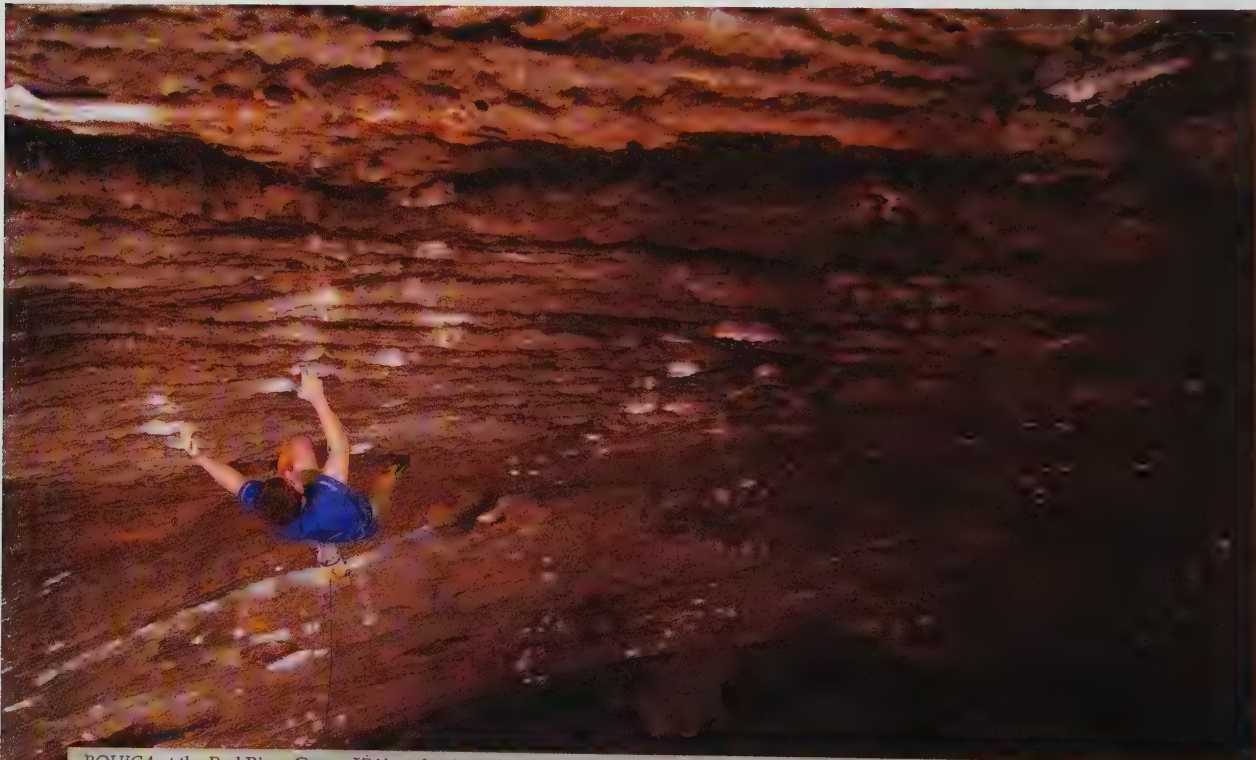
AEROBIC RESTORATION AND CAPILLARITY TRAINING

For rock climbing, the most practical method for improving capillarization in the forearms and raising the MSS is “aerobic respiration and capillarity” (ARC) training.⁴ This consists of climbing for long, sustained periods as close as possible to the MSS. This is best performed by climbing on vertical to slightly overhanging terrain that places a steady load on the forearms so that a moderate, but sustainable pump ensues for upwards of 30 minutes. (Interval training is also beneficial for improving endurance, and is discussed in [Chapter Eight: Power-Endurance](#).) In climbing, ARC training is used to:

“...I started to learn the value of recovery. With Bachar’s influence, this would often involve ‘active rest.’ Not total rest, but days of doing the activity at a much-reduced level. For a sprinter, on a rest day, you would go out and do a bit of a jog. As a climber, I would go on easy soloing circuits. It keeps coordination and flow intact.”

— Jerry Moffatt⁷

- Develop a foundation of muscular fitness to prepare for more-intense training
- Improve local forearm capillarity for improved aerobic energy metabolism
- Improve grip control (to prevent “over-gripping”) and pump-management skills
- Aid recovery from intense training and performance efforts
- Acquire and practice new movement skills and techniques



BOHICA at the Red River Gorge, KY is a classic endurance route that rewards those with a high MSS – though rated 5.13b, most of the moves are 5.11 to 5.12-, so climbers who can ARC at the 5.12- level will cruise this commanding line, just like Greg Kerzhner.  Dan Brayack

Because of the moderate intensity, ARC training gradually develops muscular fitness in all of the prime movers needed for climbing. This provides a gradual introduction to the more-intense activities that will be introduced later in the training cycle, stimulating modest adaptations to prepare the body for the upcoming additional stress. More importantly, this training establishes the base level of fitness that each climbing performance relies on, and off which other performance characteristics (strength, power, and power-endurance) build. Therefore, Base-Fitness training is an integral part of any training program.

The major goal of ARC training is to improve the forearm muscles' local ability to utilize aerobic energy metabolism, and thus reduce reliance on the other two, less-efficient processes (glycolytic and phosphagen systems). Another way to illustrate this is in terms of quantifiable climbing performance. For example, imagine a pumpy route that is composed of several 5.10 cruxes with few rests; this would be rated 5.11. A climber whose MSS is right at 5.10 (he can climb 5.10 moves uninterrupted for half an hour without getting tired) would be able to redpoint this climb easily because he would never need to rely on the less-efficient metabolic pathways. Now imagine he tries a 5.12 route that is packed with 5.11 cruxes. These will push him beyond his MSS, leading him to get pumped and possibly fail on the route. If he raised his MSS through ARC training to the 5.11 level, he would be able to climb this second route easily.

Improvement in your ability to utilize local aerobic metabolism directly enhances the ability to hold onto small holds longer, delay the onset of a pump, climb more-difficult moves while pumped, and recover from difficult moves more quickly. This is how the superstars onsight 5.14, or free a Grade VI big wall in a day; their MSS is so high that they can accomplish these feats while climbing aerobically. While aerobic climbing capacity benefits all aspects of performance, it is absolutely critical on certain types of routes, especially onsights and long, pumpy redpoints.

Proper ARC training requires careful moderation of forearm effort in order to maintain muscular exertion right at the MSS. This attention to hand-grip control is extremely valuable in difficult climbing as well, and will spell the difference between success and failure on routes of all kinds. Therefore, ARCing provides a great opportunity to hone this essential skill. The ability to manage a mounting pump throughout a climb is related to grip control, and is also essential to proper ARC training. The ARCing climber will quickly

develop the ability to pace climbing sessions, time rests, and modulate effort throughout a climb to keep the pump in control.

Another critical purpose of ARCing is to speed recovery between workouts or taxing climbing days. Blood flow through skeletal muscle is stimulated by muscle contractions, through a system called the "skeletal muscle pump" (not to be confused with the climbing "pump"). The cyclic contraction of a muscle helps to transfer blood from capillaries to the veins so it can be transported back to the lungs for re-oxygenation. This process helps remove waste products and supply the muscle with fuel and other raw materials to rebuild muscle tissue after a hard workout. Inactivity causes blood to pool, and deprives the muscle of essential nutrients. ARCing provides the perfect stimulus for increased blood flow, as well as improved recovery. This "flushing" effect can be clearly felt after a hard climbing day.

Finally, ARC training is the perfect venue for learning and practicing novel climbing movements and techniques. The relatively low level of intensity permits the climber to concentrate on the subtleties of climbing technique, and to explore novel move-



Boulder-traversing in a gym is the simplest way to perform ARC training, and provides an opportunity for learning and reinforcing new techniques.

The Lost Art of ARC

ARCing may be one of the most derided forms of climbing training because it is hard to see the connection between the effort and tangible results. Many feel that because ARCing feels “easy” it must not be helping; “no pain, no gain,” right? Not necessarily.

Early in my career as a climbing athlete (the point at which I started dedicated training), I ARC’d a lot. I did long (four-to-six-week) phases of dedicated ARC training, three sets a day, five days per week. At the time, it wasn’t clear that these sessions had any direct benefit. I was improving at climbing, but I attributed those gains to other training activities. (Part of this conclusion resulted from the routes I was climbing: They were short and bouldery, and rarely had any endurance component.) Years later, I’m convinced that the time I spent ARCing had tremendous value, and it comes down to technique. In my career, I’ve managed to climb several highly technical routes. According to others, I have relatively precise footwork, smooth, accurate movement, and little wasted effort. I didn’t start out this way; it came from hours upon hours of moderate climbing. Furthermore, this was pre-iPod, so I had nothing better to think about than dialing in my movement skills.

Fast-forward ten years, and now I climb extensively on steeper, more sustained routes, especially at the Red River Gorge, where much of the climbing is done at or very near the MSS, with brief excursions beyond the MSS. Therefore, the ability to raise my MSS through ARC training has had tremendous physical benefit, in addition to the technical benefits such as improved grip control and relaxation. A great example of the power of ARCing is the Motherlode’s premier route, *Transworld Depravity*. *Transworld* has three distinct cruxes, each slightly easier than the last, separated by more-moderate but pumpy climbing. I decided to ARC (climb below my MSS) between the cruxes, and suppress the pump until I hit each crux, where I expected the pump to build. As I finished each crux and arrived at more-moderate terrain, the skills and physical adaptations I’d developed through ARCing would enable me to recover from the crux, clear out the pump, and set up for the next crux. I had to focus intensely to stay relaxed, control the pump, and keep climbing, but the ARCing paid off, and I redpointed *Transworld* in a quick four days of work. At the Red, raising my MSS corresponded directly to an increase in high-end climbing performance.

— Mike Anderson



Mike Anderson ARCing his way up *Transworld Depravity*, 5.14a, at the Motherlode, Red River Gorge, KY.  Janelle Anderson

ments while still achieving physical training goals. The duration of ARC sets provides ample time for repetitive practice of new techniques.

When properly performed, ARC training causes three key adaptations in the muscles of the forearm:

1. Increased blood *flow* increases shear stress on the capillary walls and stimulates capillary growth. Increased blood *pressure* stimulates cross-sectional growth of blood *arterioles* (the blood vessels that feed into the capillaries). This growth can occur after as little as two weeks of training, and will improve blood flow in the forearms while climbing.¹²
2. Stimulates the release of a protein (VEGF) that initiates *angiogenesis*, or capillary growth (training at altitude can stimulate similar adaptations).
3. Stimulates increases in the quantity, size, and capacity of mitochondria in the affected muscle cells, which improves the muscles' ability to utilize aerobic respiration to the maximum extent possible, during all aspects of climbing performance.

PERFORMING ARC TRAINING

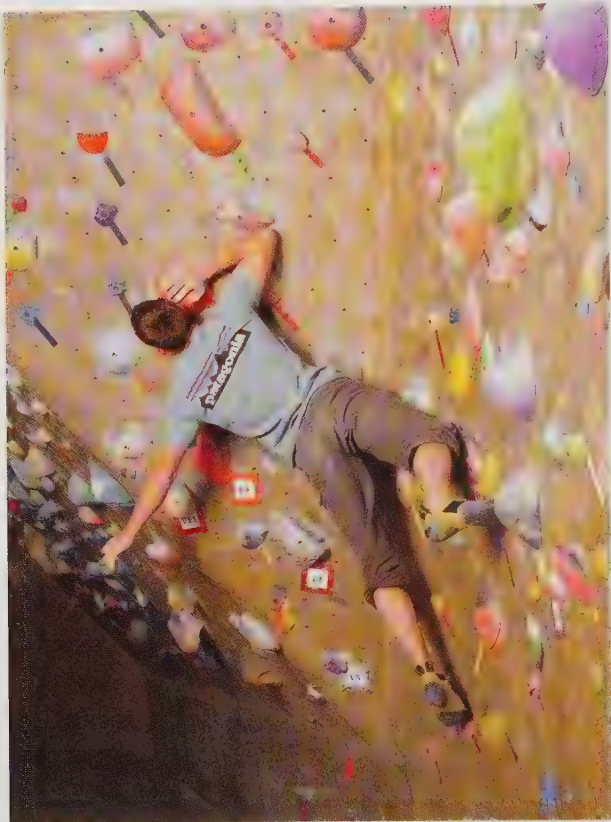
An ARC workout involves sustained climbing, usually in a climbing gym, right at the MSS. A single set typically consists of climbing, without a break, for 20 to 45 minutes straight. Up to three sets can be completed in a single ARC workout. The goal is to continuously climb at the MSS: the level of intensity that taxes the muscles of the forearms such that aerobic respiration of the muscle cells can just barely meet the energetic demands for ATP. This level of effort will stimulate desired adaptations, which are dependent on high blood flow through the capillaries. Climbing on terrain that is too difficult, above the MSS, will occlude blood flow, preventing this necessary stimulus. The volume of ARC training completed determines the level of improvement achieved, so the more ARC training, the better (barring injury).

FOCUS AND INTENSITY

Since climbers are not equipped with an “ATP consumption gauge,” we must use subjective indicators to judge the correct level of effort — mainly, the level of forearm pump. The key to performing the workout properly (thus stimulating the proper adaptations) is to modulate the difficulty of moves in order to maintain the level of muscular effort right at, or below, the MSS. Begin with more difficult climbing to build to a moderate pump, and then adjust the difficulty as needed to maintain the pump. Constant focus is necessary throughout each set to assess and control

the level of intensity. Pay close attention to the body's feedback, and adjust the terrain and hold selection as necessary. Here are some tips for maintaining proper focus and intensity:

- ARCing must be done by *feel*; with practice, the feel of a strenuous but sustainable pump will become second nature. A good ARC session should have the climber breathing heavily and lightly sweating after the first 10 minutes or so.
- Most climbers will spend the majority of each set on vertical to just slightly overhanging terrain.
- Each set should last at least 20 minutes; therefore, develop and maintain the level of pump that can be sustained for that duration. It should be as severe as is tolerable short of falling off or flailing. That said, the goal is NOT to pump off the wall just as the clock strikes midnight. With varied terrain and an assortment of holds, you should be able to modulate the grip size, angle, and movement difficulty as needed to “dial in” the perfect pump.
- If the level of pump is too great, blood flow will be occluded, preventing the beneficial adaptations stimulated by high blood flow and high blood pressure.
- Every move in an ARC set doesn't have to be of equal difficulty. It is acceptable to climb a short series of hard moves followed by an easier set of moves to recover. For example, a climber could alternate between short stints of small and large footholds, if the gym's walls are too steep to use small footholds exclusively. This works well for practicing specific movements that are slightly too hard for ARCing.
- If a severe pump develops, the moves are too difficult; if no pump, sweating, or heavy breathing develops then the moves are too easy.
- Effective training is always progressive, so expect to *slightly* increase the intensity of the ARC sessions over the course of the phase. Progress is gradual but persistent, so don't expect meteoric gains right away.
- Once the correct level of effort is determined, add in movement practice, breathing, and other skills discussed in **Chapter Three: Skill Development** to create a workout that is equal parts physical training and skill practice.



ARC sessions provide a great opportunity to discover and practice novel rest positions.

- Emphasize smaller footholds to increase climbing specificity.
- A fun challenge is to locate increasingly improbable and awkward rest positions while ARCing. Look for unlikely stems, highsteps, mantels, heel hooks, etc. (refer to **Chapter Three: Skill Development**). Avoid easy rests that unweight the arms, practicing active rests instead (those that require alternating hands and labored breathing). This practice will be much appreciated when the Performance Phase rolls around.
- In the gym, ARCing should not be limited to designated (taped) routes; utilize all the available holds and features to modulate intensity and specificity.
- ARCing can be difficult to accomplish in a crowded public climbing gym, so try to visit the gym at an off-peak time. Treadwalls or auto-belay systems can be great tools for ARCing, if available.
- If a dedicated training partner is available, consecutive up-down-up-etc. sessions on toprope can help spice up the monotony of ARCing.

- ARC-leading in a gym is one of the ultimate forms of training, creating physical, technical, and mental gains. Tie in, and up and down-climb (unclipping while downclimbing so you can climb back up a different route) on a lead wall for 30 minutes, then swap roles with the belayer. This provides critical practice for relaxed lead climbing, clipping, resting, etc., and is the perfect opportunity to “stress-proof” new movement techniques. These workouts should be scheduled late in the Base-Fitness Phase.
- ARC training is not a mutually exclusive workout; nothing has to be replaced in order to include it in a training program. It can be performed at the end of a workout, or the day after a hard workout.
- The total sum of ARCing time determines improvement, not the duration of individual sets. I.e., three sets of 30 minutes will theoretically produce the same effect as two sets of 45 minutes. Many climbers find it easier to maintain focus and intensity through a greater number of shorter sets as opposed to two mega-sets.
- Too much ARCing can lead to tendonitis in the fingers. Pay attention to signs of overuse, and plan adequate rest.

“In general, endurance work is all about volume, so if you’ve got the time, then the best place to get fit is at the crag, simply by putting the mileage in.” — Ben Moon¹³

OUTDOOR MILEAGE TRAINING

While the typical Base-Fitness workout is performed by ARCing in a climbing gym, it is beneficial to perform these workouts outside on occasion. In terms of skill development, outdoor routes are preferred, as they provide a more realistic practice environment. Throughout the training season, indoor ARC workouts may be substituted with Outdoor Mileage (OM) days in which six to twelve “moderate” routes or pitches are climbed out at the crags. These routes should be approximately one to two full number grades below the climber’s maximum onsight ability; they should not require rehearsal or “all-out” efforts to complete, but they should be moderately fatiguing — just like ARCing. Apply the guidelines of ARC training (described previously) to these routes, concentrating on

“A lot of the time when you’re climbing, John [Bachar] explained, you really want to be letting go of holds, not gripping them harder. Find ways of applying just the right amount of force to a hold, he suggested, instead of wasting lots of energy. Relax the entire body. Flow.” — Jerry Moffatt⁷

details such as breathing, grip control, movement, etc. Try to create continuous ARC “sets” of 20 to 45 minutes, which may require fast climbers to climb multiple pitches back-to-back. Think of OM workouts as simply ARCing on a rope, outside. Finally, OM days should be fun. This is the opportunity to spend a stress-free day at the crag with friends, ticking off enjoyable classics or conquering that multipitch route you’ve been meaning to climb.

SKILL DEVELOPMENT WITH BASE-FITNESS TRAINING

The moderate intensity of Base-Fitness training makes it ideal for practicing climbing movement, including acquiring new skills and rehearsing and refining existing skills. **Chapter Three: Skill Development** covers this topic in detail, suggesting many drills that can be accomplished to develop and rehearse critical climbing skills. Many of these should be performed during ARC workouts, but there are a few guidelines to keep in mind:

- Correctly performing technique drills requires concentrating on the drill itself, but don’t forget to pay attention to the level of physical exertion. Maintain proper intensity to achieve the correct ARC training effect. Climbers new to ARC training should wait to incorporate technique drills for a few sessions until they have found suitable terrain and determined the right level of effort to reach their MSS.
- It is best to concentrate on only a few techniques and/or drills in a given session. This allows the climber to remain mindful of the key focus areas of each drill and perform them correctly.
- Specific techniques and drills should be progressively revisited in subsequent ARC sessions throughout a season, as suggested in the calendar in the Figure below.
- A major technical goal of ARC training is to learn to climb relaxed and remain calm, regardless of the mounting fatigue or tactical errors on a route. Listening to mellow, soothing music while ARCing may help encourage this mindset.

SCHEDULING BASE-FITNESS TRAINING

A typical Base-Fitness workout is two or three 30-minute continuous climbing sessions, or “sets,” with 10 minutes of rest between each set (bring your MP3 player!). Again, the total duration of MSS climbing is more important than how the sets are split up; therefore, aim for a total of 60 to 90 minutes of climbing time in an ARC or OM workout, with at least 20 min-

Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
						Sa Day 1 *Outdoor Mileage (OM), 6 mod pitches *Skill acquisition
Su 1 *OM, 4 mod pitches *Skill acquisition	Mo 3 *Optional Aerobic Exercise (OAE)	Tu 4 *ARC 1: 2 x 20 min *Skill acquisition	We 5 *OAE	Th 6 *ARC 2: 2 x 25 min *Skill acquisition	Fr 7 *OAE	Sa 8 *OM, 7 mod pitches *Skill acquisition
Su 9 *OM, 5 mod pitches *Skill acquisition	Mo 10 *OAE	Tu 11 *ARC 3: 2 x 30 min *Skill practice	We 12 *OAE	Th 13 *ARC 4: 2 x 30 min *Skill practice	Fr 14 *OAE	Sa 15 *OM, 8 mod pitches *Skill practice
Su 16 *OM, 6 mod pitches *Skill practice	Mo 17 *OAE	Tu 18 *ARC 5: 3 x 25 min *Skill practice	We 19 *ARC 6: 3 x 25 min *Skill practice	Th 20 *ARC 7: 2 x 25 min *Stress-proofing	Fr 21 *OAE	Sa 22 *OM, 10 mod pitches *Stress-proofing
Su 23 *OM, 8 mod pitches *Stress-proofing	Mo 24 *OAE	Tu 25 *ARC 8: 3 x 30 min *Stress-proofing	We 26 *ARC 9: 3 x 30 min *Stress-proofing	Th 27 *ARC 10: 2 x 30m *Stress-proofing	Fr 28 *OAE	Sa 29

A typical Base-Fitness schedule for a novice climber, consisting of indoor ARC sessions and Outdoor Mileage sessions. Notice that the volume of ARC training increases throughout the phase, and Skill Development activities are specified and also progress from acquisition, to practice, to stress-proofing.

Key:
 ARC – Aerobic Restoration and Capillarity Training
 OM – Outdoor Mileage
 OAE – Optional Aerobic Exercise

utes of continuous climbing per set. A typical four-week Base-Fitness training schedule for a novice climber is shown on p101.

ARC training is not very intense, so the climber does not need to be completely rested, or “fresh,” to ARC. Therefore, ARC workouts can be scheduled the day after intense training days (such as strength work-

outs), though typically you’d perform at a reduced volume (two 20-to-30-minute sets, or one 45-minute set). This works particularly well when two rest days are taken between intense workouts. For example, a typical **three-week** Strength Phase schedule for a novice climber that includes optional, supplemental Base-Fitness workouts is:

Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
Su 23	Mo 24	Tu 25	We 26	Th 27	Fr 28	Sa 29
	<i>Last Week of Base-Fitness Phase</i>					*Hangboard (HB) 1 *Supplemental Ex
Su 30 *Optional OM. 4p *OAE *Skin care	Mo 31 *OAE	Tu 32 *HB 2 *SE	We 33 *Opt ARC: 2x20 min *OAE *Skin care	Th 34 *OAE	Fr 35 *HB3 *SE	Sa 36 *Optional OM 4p *OAE *Skin care
Su 37 *OAE	Mo 38 *HB 4 *SE	Tu 39 *Opt ARC: 2x20 min *OAE *Skin care	We 40 *OAE	Th 41 *HB5 *SE	Fr 42 *Opt ARC: 2x20 min *OAE *Skin care	Sa 43 *OAE
Su 44 *HB6 *SE	Mo 45 *Opt ARC 1 x 30min (Begin Tapering) *Skin care	Tu 46	We 47 *HB7 *SE	Th 48 *Skin care	Fr 49	Sa 50

A typical Strength Phase schedule for a novice climber, which includes supplemental Base-Fitness training (indoor ARC sessions and Outdoor Mileage sessions). These workouts are optional during the Strength Phase, so they are shown in grey text. Notice that the volume of ARC training is lower than it is during the Base-Fitness Phase, and tapers near the end of the phase.

Key:

ARC – Aerobic Restoration and Capillarity Training
OM – Outdoor Mileage
OAE – Optional Aerobic Exercise
HB – Hangboard Exercise

In the above example, the Base-Fitness workouts (ARC or OM) can be completed the day after strength training because they are not very intense, so the fatigue of Day 32 will not impair Day 33’s ARC workout. Further, because Day 34 is a rest day, there is plenty of time for the body to recover from any lingering fatigue caused by the ARC workout on Day 33, before strength training again on Day 35. As an added benefit, the ARC workout will aid recovery from Day 32’s strength workout by encouraging blood flow through the climbing-relevant muscles, improving the climber’s performance on Day 35.

In addition to entire workouts consisting of multiple ARC sets, it can be beneficial to incorporate one or more ARC sets into other workouts or climbing days. For example, 20 to 30 minutes of ARCing may be performed at the beginning of a strength workout as part of the warm-up, or a 30-minute ARC set may be done at the end of a climbing day as a cooldown (many climbers end a day with “laps” on a moderate route, which creates a similar effect). This “modularity” is a great feature of ARC training; ARCing can thus be squeezed into a training program in many different ways, without interfering with other forms of training.

In the periodic Rock Prodigy program, dedicated ARCing is scheduled at the start of each climbing season in a two-to-four-week phase that prepares the climber for the more-intense training activities to come. After this initial phase, ARC workouts can be included occasionally within the Strength and Power-Endurance phases (at a reduced volume, as described above). ARC workouts performed in the latter phases are supplemental and should be reduced or eliminated if they interfere with the primary workouts of that phase. Interference might include inadequate recovery between workouts, signs of impending overuse injury, excessive skin wear, or insufficient time to accommodate all of the weekly training activities.

SUMMARY

- Building a training base prepares the climber for more-intense training and establishes the baseline for further muscular performance.
- Muscle function is complex, with many redundant adaptable systems that can be trained, as desired, to permit sustained activity and/or increased force production.
- There are three metabolic energy pathways that the muscles use to varying degrees to provide fuel for muscular contractions. The more the muscles use aerobic respiration, the longer climbing can be sustained.
- Muscle fatigue is specific to the activity, so endurance training must be specific to the climber's goals.
- Because of the importance of aerobic respiration, all climbing disciplines will benefit from Base-Fitness training, which improves mitochondrial density and performance.
- A forearm pump results from occluded blood flow in the forearms, limiting the delivery of fuel and removal of waste. Increasing strength, controlling grip effort while climbing, and improving local forearm capillarity will all help to delay the onset of a pump.
- Local muscular capillarity and aerobic efficiency are increased through aerobic restoration and capillarity (ARC) training.
- ARC training must be performed at the correct intensity to achieve the desired results. ARCing above the MSS will occlude blood flow, preventing the necessary training stimulus.
- ARC training has several other benefits, including facilitating skill development, accelerating recovery, and improving grip control.



Climbing long, sustained multipitch routes, such as *La Fiesta del Biceps*, 7a, at Riglos, Spain, is a brilliant way to perform Base-Fitness training.

 Andrew Burr